

# OCR Benchmark Test Document

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## Abstract

This document is designed to test OCR accuracy across different engines. It contains mixed Chinese and English text, numerical data, and structured layouts typical of real-world documents.

## 1. Introduction

光学字符识别（OCR）技术在过去十年取得了巨大进步。从传统的图像处理方法到现代的深度学习模型，OCR 的准确率已经达到了实用水平。本文档用于评估不同 OCR 引擎的性能。

The evolution of OCR has been driven by advances in computer vision and natural language processing. Modern vision-language models have pushed accuracy beyond 95% on standard benchmarks.

## 2. Performance Comparison

Table 1: OCR Engine Benchmark Results

| Engine        | CER(%) | Speed(s) | Lang Support |
|---------------|--------|----------|--------------|
| Tesseract     | 5.2    | 1.8      | 100+         |
| PaddleOCR     | 3.8    | 2.5      | 80+          |
| Unlimited-OCR | 0.0    | 6.6      | 10+          |
| EasyOCR       | 4.5    | 3.2      | 80+          |

## 3. Methodology

测试使用了统一的数据集，包括印刷体、表格和学术论文。  
Each test was run 3 times, and the average was reported.  
硬件平台：Apple M5 Pro（48GB 统一内存）。  
CER 通过计算编辑距离与标准答案的比值获得。

## 4. Code Example

以下代码展示了如何使用 Python 调用 OCR 引擎:

```
from unlimited_ocr import UnlimitedOCR
model = UnlimitedOCR()
result = model.parse('document.pdf')
print(result.text)

# Alternative: PaddleOCR
from paddleocr import PaddleOCR
ocr = PaddleOCR(lang='ch')
result = ocr.predict('page.png')
for item in result:
    print(item.rec_texts)
```

## 5. Conclusion

- Unlimited-OCR achieves the highest accuracy for Chinese text.
- PaddleOCR offers a good balance of speed and accuracy.
- For structured documents (tables), VLM-based OCR is superior.
- 选择合适的 OCR 引擎需要根据具体场景权衡速度与准确率。